

Project Sample	
Student Name	xxx
Student ID	123456789

You shall:

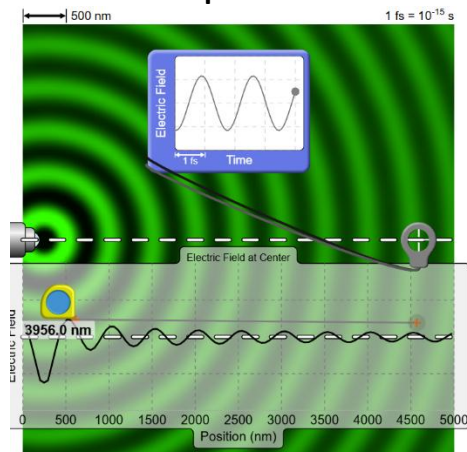
- Capture the screens of your measurements as your data and include in HW report.
- Compare your results with the textbook ones and explain the reasons for any discrepancies.

Assignment 1: Light wave

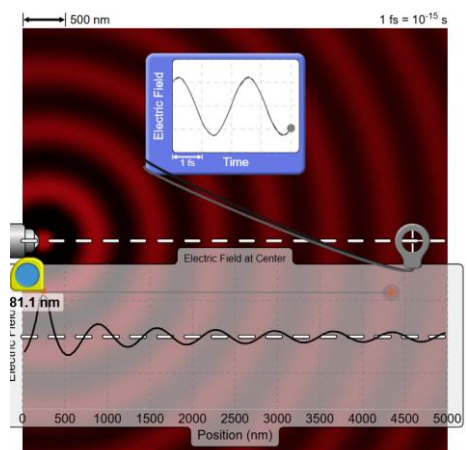
Simulation: https://phet.colorado.edu/sims/html/waves-intro/latest/waves-intro_all.html

Tasks: Choose Light

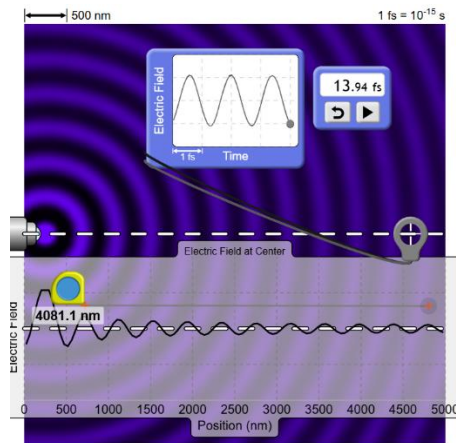
- (1) Measure the frequency and wavelength of red, green and blue light. (think about how to reduce measurement errors).
- (2) Calculate the speed of each color.



- 8 X wavelength = 3956.0 nm,
- 2Xperiod = 3.5 fs,
- speed = wavelength X frequency
= wavelength / period = 494.5
nm / 1.7 fs = 290.9 nm/fs =
290.9 x 10⁶ m/s

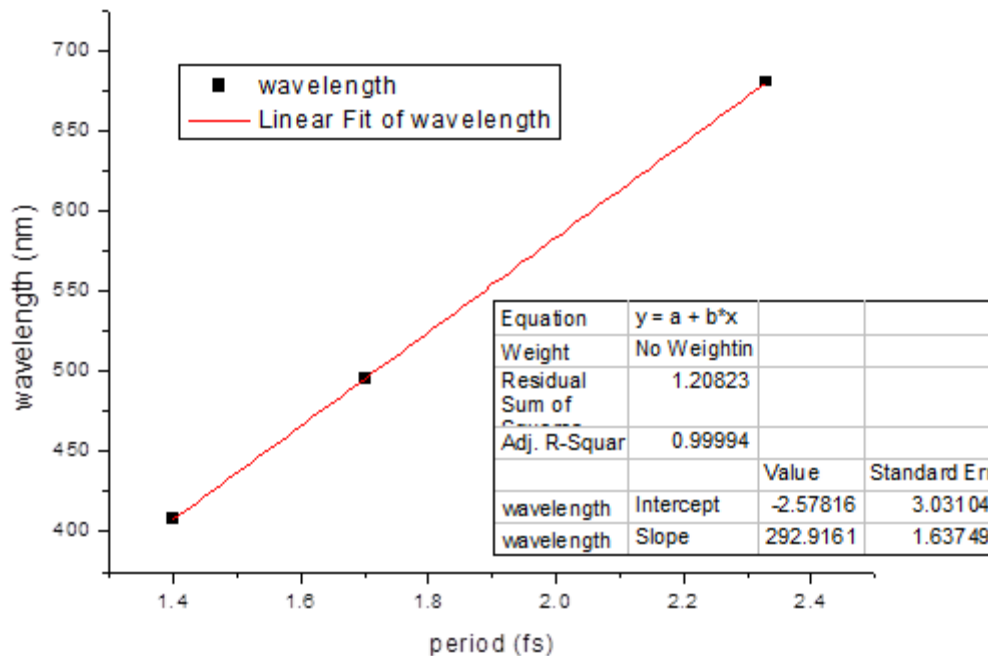


- 6 X wavelength = 4081.1 nm,
- 1.5 X period = 3.5 fs,
- speed = wavelength X frequency
= 680.2 nm / 2.3 fs = 291.9nm/fs
= 291.9 x 10⁶ m/s



- $10 \times \text{wavelength} = 4081.1 \text{ nm}$,
- $\text{period} = 1.4 \text{ fs}$,
- $\text{speed} = \text{wavelength} \times \text{frequency}$
 $= 408.1 \text{ nm} / 1.4 \text{ fs} = 291.5$
 $\text{nm/fs} = 291.5 \times 10^6 \text{ m/s}$

(3) Plot a graph showing wavelength against period (1/frequency)



(4) What conclusion do you draw based on (3)?

Speed of light = 292.9 (nm/fs)

(5) Suggest how to reduce the measurement error.

Measure more colors.

Count longer time to measure frequency.